

Real-time Hand-object Occlusion for Augmented Reality Using Hand Segmentation and Depth Correction

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Figure 1: Hand-object occlusion effects achieved by the proposed method under different hand states: (a)-(b) single hand; (c) hand with an object; (d) both hands

ABSTRACT

Hand-object occlusion is crucial to enhance the realism of Augmented Reality, especially for egocentric hand-object interaction scenes. In this paper, a hand segmentation-based depth correction approach is proposed, which can help to realize real-time hand-object occlusion. We introduce a lightweight convolutional neural network to quickly obtain real hand segmentation mask. Based on the hand mask, different strategies are adopted to correct the depth data of hand and non-hand regions, which can implement hand-object occlusion and object-object occlusion simultaneously to deal with complex hand situations during interaction. The experimental results demonstrate the feasibility of our approach presenting visually appealing occlusion effects.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Computing methodologies—Computer graphics—Image manipulation—Image processing

1 INTRODUCTION

Augmented Reality (AR) technology has been applied to many fields such as medicine and entertainment. Hand-object interaction is an important part of AR, especially for egocentric AR headset applications. For AR hand-object interaction applications, hand-object occlusion is crucial. Incorrect hand-object occlusion seriously affects the user's perception and manipulation of virtual objects, resulting in severe damage to the user's sense of AR experience.

Due to the highly dynamic nature of the hand-object interaction scenes, handling hand-object occlusion is not a trivial task. In recent years, many researchers have devoted themselves to the research of hand-object occlusion. Feng et al. [2] used a pre-customized hand model combined with Leap Motion to track the joint information of the hand, and obtained the hand occlusion mask by comparing its depth with the virtual object. Later, they [3] proposed an occlusion-aware deep learning system that jointly performs hand poses estimation and mask segmentation, which was more automatic than [2] and improved the accuracy of the hand model. Battisti et

al. [1] adopted skin detection to segment the hand, used the distance transform algorithm to associate depth values obtained by Leap Motion to hand pixels, and then performed depth comparison to realize hand-object occlusion. Tang et al. [7] proposed a deep neural network to directly predict hand-object occlusion without depth acquisition, but its generalization capability to virtual objects was constrained by the dataset.

As mentioned above, deep learning has been attempted to be applied to occlusion handling [3, 7]. Although neural networks were used for different purposes, they have shown their feasibility in achieving occlusion handling. In this paper, a real-time hand-object occlusion method based on RGBD camera was proposed, which used a compact deep neural network to quickly extract the hand segmentation mask, and adopted different strategies to correct the depth data of different regions based on the mask. The hand-object occlusion effects demonstrate the feasibility of the proposed method.

2 METHODS

2.1 Hand Segmentation Model

Hand is a key component of hand-object occlusion in AR applications. In order to improve the quality of depth data of hand and non-hand regions, it is essential to extract the hand mask quickly and accurately. To this end, we introduce a lightweight U-Net [6] network, which takes a color image as input and outputs the estimated hand binary mask. In the encoder part, we use MobileNetV3-Large [4], each bottleneck of which adopts depthwise separable convolutions, and the inverted residual structure in each block can effectively reduce the loss of high-dimensional information. Therefore, MobileNetV3-Large can not only reduce the amount of parameters and computational cost, but also have strong feature extraction ability. When using it as the encoder, we keep the first convolutional layer and the subsequent 15 bottlenecks unchanged, and remove the last four layers to meet the needs of the hand segmentation task. In the decoder part, before each upsampling, we introduce the CBAM [8] module, a channel and spatial attention mechanism to make the network better focus on hand features.

2.2 Depth Correction Based on Hand Mask

Before performing depth map correction, we first need to align the depth map to the color image as a result of the different parameters of the depth sensor and the color sensor of the RGBD camera. Aiming at the defects of the depth map, we then use the hand mask segmented by the neural network to perform depth correction in

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different strategies for hand and non-hand regions to obtain more accurate depth map. The specific details of the algorithm are as follows.

2.2.1 Hole Filling

Hand depth data is critical for hand-object occlusion, so the holes in the hand region need to be filled as much as possible. According to the hand mask, for each hand pixel with zero depth, the valid depth with the closest Euclidean distance in the neighborhood and also located in the hand mask is used for filling. By choosing an appropriate neighborhood size, we can achieve the complete filling of the hand holes. For each zero depth in the non-hand region, we define an $N \times N$ pixel filter. According to the Chebyshev distance ($d_{12} = \max(|x_1 - x_2|, |y_1 - y_2|)$) between the neighboring pixels and the center, the neighborhood is divided into multiple layers, and the corresponding count threshold is set for each layer. Count the number of valid depths in the non-hand region of each layer. If the threshold for either layer is broken, the statistical mode of all counted depths will be applied to the central zero depth.

2.2.2 Depth Correction within and around Hand

Depth maps generated by low-cost depth sensors are generally noisy, especially in areas near depth edges. In hand-object occlusion, the depth data within and around hand is a key factor and needs to be corrected accurately. For depth data within hand, we first calculate the mean depth of the hand according to the mask. Based on the mean, we judge whether the depth of each hand pixel is valid through the threshold range. For each invalid depth, we replace it with the mean of the valid depths of the hand pixels in the neighborhood. As a result of the depth map alignment error and the hole filling described above, there may be depths around the hand that are similar to the hand edges. For these pixels, we first morphologically dilate the hand mask M with a large structural element to get M_1 , and then morphologically dilate M_1 with a small structural element to get M_2 . The intersection of M_1 and M is the region C that depth data needs to be corrected, while the intersection of M_2 and M_1 is the target region T . For each depth of C , if it is similar to the hand edges' depth, we use the depth with the closest Euclidean distance in T to replace it, thus realizing the depth correction around hand. Fig. 2 shows the depth map obtained after this correction step, which shows that the depth data within and around hand can be significantly improved.

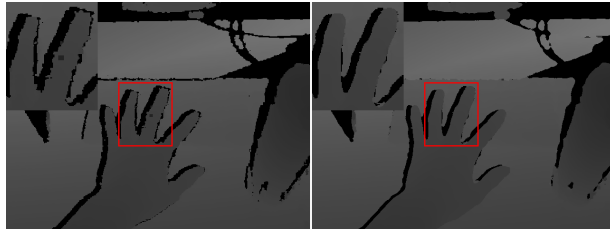


Figure 2: Raw depth map (left) and after correction by 2.2.2 (right).

2.2.3 Mask-based Joint Bilateral Filtering

Because of the complex states of the hand in hand-object interaction scenes, there may be objects in user's hand. To cope with such situations, we need to ensure that the non-hand region where the objects are located also has relatively accurate depth data to achieve object-object occlusion. Joint bilateral filtering has the characteristics of removing noise while preserving edge information, which has been confirmed by [5] to have a good effect in correcting depth maps. Since there is no correlation between the depth data of the hand and non-hand regions, we adopt joint bilateral filtering based on the hand mask. It takes color image as the guide image. When calculating the Gaussian weights, each pixel in the neighborhood only contributes to the central pixel belonging to the same region. After mask-based joint bilateral filtering, the depth correction is realized.

3 RESULTS

In the experiment, an Intel RealSense SR300 was used as the input of the hand-object occlusion pipeline. We implemented and trained the hand segmentation model in PyTorch with BCEloss as the loss function. The Multipliers parameter of MobileNetV3-Large was set to 1.0, so that the model has fast segmentation speed and high accuracy. We exported the trained model and used the C++ interface LibTorch to call the model directly. The entire depth correction algorithm was implemented on GPU by CUDA, and OpenGL depth test was used to implement occlusion. For 640×480 inputs, on a system with an Intel i7-11700 CPU and an NVIDIA GeForce RTX 3060 GPU, the frame rate of the hand segmentation model is faster than 60FPS, and the frame rate of the whole hand-object occlusion pipeline can reach 37FPS.

The hand-object occlusion effects achieved by the proposed method are shown in Fig. 1, which proves that the proposed method can present accurate and visually appealing occlusion effects both with and without objects in the hand. Note that we can even achieve both hands occlusion when user's hands are not overlapping.

4 CONCLUSION

In this paper, we proposed a novel approach to handle hand-object occlusion in real time by combining hand segmentation and depth correction. A lightweight deep neural network was proposed to segment the hand in real time and different strategies of depth correction were performed based on hand mask. The results demonstrate that the depth maps corrected by the proposed method are able to achieve visually appealing hand-object occlusion effects. Future work will focus on the distinction between right and left hand in order to improve the robustness of hand-object occlusion when user's hands overlap. In addition, we will refine the experiment to provide qualitative and quantitative results to further discuss the performance of the proposed method in various aspects.

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